Check for updates

A Cost-Driven Design Methodology for Additive Manufactured Variable Platforms in Product Families

Xiling Yao
Singapore Centre for 3D Printing,
School of Mechanical and Aerospace Engineering,
Nanyang Technological University,
50 Nanyang Avenue,
Singapore 639798, Singapore
e-mail: yaox0006@e.ntu.edu.sg

Seung Ki Moon[1]
Assistant Professor
Singapore Centre for 3D Printing,
School of Mechanical and Aerospace Engineering,
Nanyang Technological University,
50 Nanyang Avenue,
Singapore 639798, Singapore
e-mail: skmoon@ntu.edu.sg

Guijun Bi
Singapore Institute of Manufacturing Technology,
71 Nanyang Drive,
Singapore 638075, Singapore
e-mail: gjbi@simtech.a-star.edu.sg

Additive manufacturing (AM) has evolved from prototyping to functional part fabrication for a wide range of applications. Challenges exist in developing new product design methodologies to utilize AM-enabled design freedoms while limiting costs at the same time. When major design changes are made to a part, undesired high cost increments may be incurred due to significant adjustments of AM process settings. In this research, we introduce the concept of an additive manufactured variable product platform and its associated process setting platform. Design and process setting adjustments based on a reference part are constrained within a bounded feasible space (FS) in order to limit cost increments. In this paper, we develop a cost-driven design methodology for product families implemented with additive manufactured variable platforms. A fuzzy time-driven activity-based costing (FTDABC) approach is introduced to estimate AM production costs based on process settings. Time equations in the FTDABC are computed in a trained adaptive neuro-fuzzy inference system (ANFIS). The process setting adjustment's FS boundary is identified by solving a multi-objective optimization problem. Variable platform design parameter limitations are computed in a Mamdani-type expert system, and then used as constraints in the design optimization to maximize customer perceived utility. Case studies on designing an R/C racing car family illustrate the proposed methodology and demonstrate that the optimized additive manufactured variable platforms can improve product performances at lower costs than conventional consistent platform-based design. [DOI: 10.1115/1.4032504]

Keywords: additive manufacturing process, cost estimation, platform optimization, product family design

1 Introduction

AM represents a collection of manufacturing processes that produce parts by bonding raw materials in a layer-by-layer manner [1]. Compared to traditional subtractive and formative manufacturing processes, AM has the remarkable benefit of enabling customized products to be designed and manufactured rapidly with more design freedoms [2]. Therefore, AM has the potential to be applied in product family development, where diversified customer requirements need to be satisfied at low costs.

Although the design of an additive manufactured part can be adjusted without tooling or fixturing changes as those in traditional processes, too much deviation from the reference part design may still introduce high cost increments due to the need of making significant process setting adjustments, such as planning new support structures and printing orientation or speed. In worse cases, additional trial-and-error runs may be required to validate new process settings for the adjusted part design, which result in undesired extra costs. In this paper, we introduce the concept of an additive manufactured variable product platform. Unlike the conventional definition of a platform that remains consistent across different products in a family [3], an additive manufactured variable platform allows design variations based on a given reference part. However, the adjustment of an AM product platform and its corresponding process setting platform are constrained within a bounded range or "FS" as defined in this paper, so that undesired high cost increments will not be incurred.

Conventional product families implemented with consistent platforms have the drawback of performance compromise due to strict commonality in platform designs [3,4]. In this research, we aim to compensate the performance compromise by utilizing the design freedom of additive manufactured product platforms while controlling AM production costs at the same time. To achieve this goal, we propose a cost-driven methodology to optimize the design of a variable platform within a cost limit. We adapt an FTDABC approach and a Mamdani model to establish the interrelationships between AM production costs, AM process settings, and variable platform's design parameters. When the proposed cost-driven design methodology is applied to additive manufactured variable platforms, the boundary of the process setting adjustment's FS is searched by solving a multi-objective optimization problem that maximizes design freedoms and minimizes AM cost increments. The FS boundary is then used as a constraint in the variable platform design optimization process which maximizes part-worth customer utilities. Case studies on designing an R/C racing car family are used to demonstrate the usefulness of the proposed methodology. The results show that implementing additive manufactured variable platforms in the product family has improved performances at lower costs compared to the conventional consistent platform-based design.

This paper is organized as follows. Section 2 reviews previous research in AM costing and platform-based product family design approaches. Section 3 introduces the concept of additive manufactured variable platforms. Section 4 describes the proposed cost-driven platform design methodology that explores AM-enabled design space with controlled costs. Section 5 describes case studies to illustrate the proposed methodology. Section 6 summarizes this research and recommends future research directions.

[1]Corresponding author.

Contributed by the Design for Manufacturing Committee of ASME for publication in the Journal of Mechanical Design. Manuscript received February 4, 2015; final manuscript received January 6, 2016; published online February 5, 2016. Editor: Shapour Azarm.

 Copyright © 2016 by ASME

from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnljTWeENb774MKM7HJ-nZUW6TxtlJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeBQw by Universita Di Pisa user on 25 November

2 Literature Review

2.1 Costing of AM Processes. A variety of AM processes have been developed, which share the same general process chain [1]. Among different types of AM processes, selective laser melting (SLM) is one of the most promising techniques for fabricating functional metal components with high speed and quality [5]. In this paper, the SLM is chosen as an example to illustrate the proposed methodologies and the case studies. AM process settings have significant impact on the cost of additive manufactured parts. For example, Cheng et al. [6], Canellidis et al. [7], and Strano et al. [8] investigated the effect of build orientation on build time, energy consumption, and the amount of required support materials. Key SLM parameters, such as laser power, scan speed, and hatch spacing, also influence AM production costs [9]. Hopkinson and Dickens [10] analyzed the unit costs of additive manufactured parts and compared them with those in a traditional manufacturing process (injection molding). Ruffo et al. [11] developed an AM cost model that consisted of direct costs and indirect costs. Atzeni and Salmi [12] introduced the cost structure specifically in metal components fabrication. All three studies [10–12] suggested that AM would have economic advantage over traditional manufacturing techniques in small to medium batch production. A generalized lifecycle cost (LCC) analysis on additive manufactured products was conducted by Lindemann et al. [13,14]. Yim and Rosen [15] proposed an early stage AM process setting selection method based on build time and estimated costs. Williams et al. [16] calculated an additive manufactured product family's cost under different process settings. In the above article, product variant designs were assumed given, while in this research, we consider the situation where designers may need to vary process parameters for different product variants whose design variables are yet determined until the final optimization is carried out.

The literature review shows that AM costing approaches have focused on estimating absolute cost values; however, they lack the capability to predict cost changes as an effect of relative AM process setting adjustments and part design variations. In this research, we adapt a time-driven costing method that relates AM cost changes with time consumption changes which are further mapped to part designs variations. Furthermore, the proposed costing method is capable to estimate uncertainties and fluctuations in cost terms through the adaption of fuzzy inference systems.

2.2 Platform-Based Product Family Design. A product family is composed of multiple product variants, and a product platform is a common part shared by more than one product variant in the family [3]. Cost savings and performance improvements are the major objectives in optimizing product family designs, while these two objectives usually conflict with each other [17,18]. Chowdhury et al. [19] developed the comprehensive product platform planning (CP^3) framework, in aim of maximizing performances and minimizing costs by identifying optimal platforms, design parameters, and production volumes. Moon et al. [4] applied multi-objective particle swarm optimization to increase both commonality and product variant performance in a product family. In the work of Jiao et al. [20], platform features were chosen to increase customer perceived benefits constructed from part-worth utilities. The concept of flexible platforms was proposed by Suh et al. [21] to allow a platform module to change over time in response to uncertainties of dynamic markets. The impact of part design changes on a product family's LCC was discussed by Wang et al. [22], which concluded that a higher design variety level would increase the cost. In the recent research on concurrent design of product families and manufacturing processes [23], a mathematical approach was introduced for joint optimization of component geometries and a reconfigurable assembly system.

Existing product family design methods rely on the assumption that identical modules are shared as platforms in a family, which has caused the drawback of performance compromise [3]. Motivated by the enhanced design freedom in AM, we hypothesize that AM enables a new platform strategy that can improve product family performance at lower or similar costs compared to conventional strategies. In this research, a novel concept of a variable platform is proposed. A cost-driven design methodology is developed for optimizing additive manufactured variable platform modules, in which the proposed costing method is applied to control and limit AM production costs.

3 Additive Manufactured Variable Product Platforms

Conventional platform design is consistent across a product family, with identical features and design parameter values shared by multiple product variants [24]. Such a platform module is defined as a *consistent platform* in this research. Due to the layer-by-layer manner of material fusion, AM processes do not involve sunk costs in tooling and fixturing like what traditional processes have, which is an advantage that enables more design freedom in customizing parts in product families. Therefore with AM processes deployed for production, we propose the concept of a *variable platform* to allow platform module itself to be varied and individually optimized, with the aim of compensating the performance compromise in consistent platform-based product families. An instance of the variable platform in a particular product variant is defined correspondingly as a *platform variant*.

However, part design changes may lead to changes in AM production costs. Therefore, a variable platform's design variations are constrained within an FS, which is to be identified in this research, so that AM production costs can be controlled and limited during the optimization of variable platforms.

4 The Cost-Driven Design Methodology for Variable Platforms

The overall procedure of the proposed cost-driven design methodology with the corresponding data flow is summarized in Fig. 1. Fundamental concepts of AM process setting platforms and variants are defined in Sec. 4.1. Sections 4.2–4.6 discuss each step of the proposed methodology as shown in Fig. 1.

4.1 Linking AM Costing With Product Family Design Through AM Process Setting Platforms and Variants. AM process settings for a variable platform are allowed to be varied in different platform variants. A set of AM process settings for producing the reference platform design $\boldsymbol{X}_R$ is defined as the process setting platform (PcP). For each individual platform variant design $\boldsymbol{X}_{\mathrm{PVi}}$, a corresponding process setting variant (PcV) is obtained by adjusting the PcP, such as fine-tuning the build orientation and replanning the support pattern. However, such adjustment operations may introduce extra cost increments due to labor, resource, and time consumptions, which need to be controlled and limited. Cost increments (CostInc) due to process settings adjustment are a function of PcV, formulated as

$$\mathrm{CostInc}_{\mathrm{PcV}} = \mathrm{CostInc}(\mathrm{PcV}) \tag{1}$$

$$\mathrm{PcV} = \mathrm{PcP} + \{\Delta\mathrm{Setting}\} \tag{2}$$

$$\mathrm{CostInc}(\mathrm{PcV}) \leq \varepsilon \quad (\varepsilon \geq 0) \tag{3}$$

where $\{\Delta\mathrm{Setting}\}$ represents AM process setting adjustment terms based on PcP and ε is the maximum allowable value of cost increments specified by the designer.

In this research, adjustments from the PcP to a PcV are constrained within an FS, defined as the maximum allowable range of process setting adjustments ($\mathrm{FS} = \{\Delta\mathrm{Setting}\}_{\mathrm{allowable}}$), so that AM production cost increments will not exceed the maximum allowable amount. The identified FS's boundary is then mapped to variable platform module's design parameters in a later step, which is used in the design optimization process for variable platform modules.

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnljTWeENb774MKM7HJ-nZUW6TxtlJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeQw by Universita Di Pisa user on 25 November

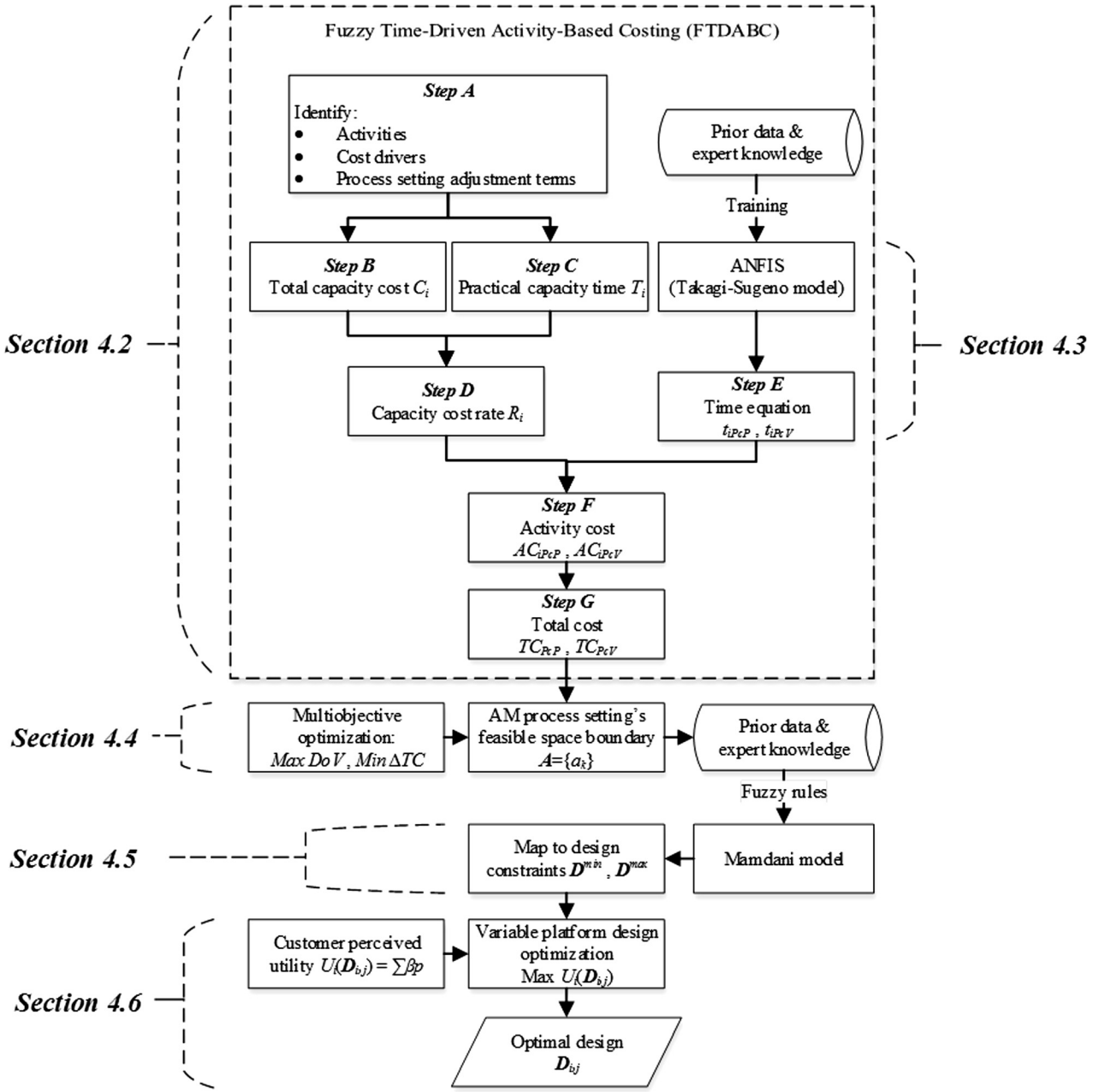


Fig. 1 Overview of the proposed cost-driven design methodology and data flow

4.2 AM Production Cost Estimation Using the FTDABC. The standard TDABC method was first created by Kaplan and Anderson [25] to calculate organizational cost in various activities. In this research, an FTDABC method is developed based on the standard TDABC to explore the relationship between AM cost increments and AM process setting adjustments. In the previous research, the fuzzy set theory has been used to model uncertainties in the total capacity cost [26] and practical capacity time [27], respectively. In the proposed FTDABC, in addition to the above two types of values, fuzziness is applied (via a trained ANFIS) to formulate uncertain time equations that are the major variables in determining total AM production costs [25].

The detailed procedure of the proposed FTDABC in AM production is explained below from steps A to G following the sequence as shown in Fig. 1. Some terminologies and notations are adapted from the standard TDABC approach [25,26].

Step A. Identify activities, cost drivers, and corresponding process setting adjustment terms in AM production

As indicated in Fig. 1, the first step of the FTDABC method is to identify activities, cost drivers, and process setting adjustment terms in producing additive manufactured variable platforms. Supposing the SLM process as an example, two main activities and their corresponding cost drivers and process setting adjustment terms are shown in Fig. 2. Each activity is explained as follows:

Activity 1: Data preparation. Prior to the fabrication, digital models of parts are processed in computer-aided manufacturing (CAM) software of the AM machine. Whenever a design is changed, the support patter, build orientation, and part position on the build platform may need to be replanned. The manpower and computer resource utilization required to make the modifications constitute the cost drivers in activity 1.

Activity 2: Build process. During the fabrication, AM process parameters may be fine-tuned to meet geometry or property requirements in different platform variant modules. Changes of energy, gas, and material consumption as well as manpower requirements constitute the cost drivers in activity 2.

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5fTPnlj7WeENb774MKM7HJ-nZUW6TxtdJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeBQw by Universita Di Pisa user on 25 November

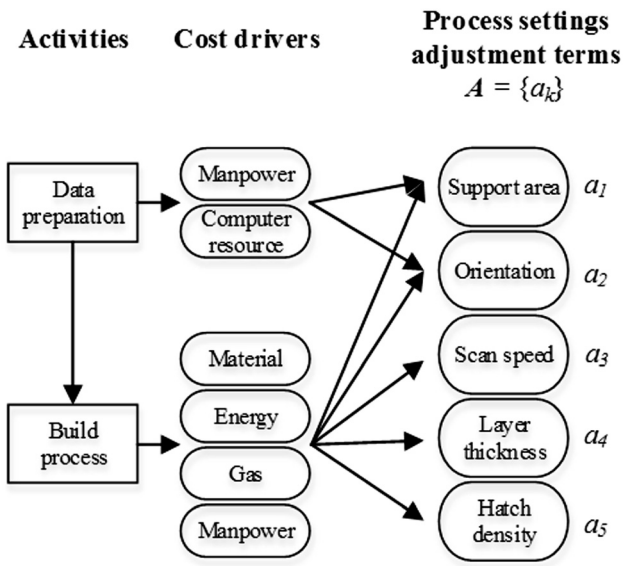


Fig. 2 Activities, cost drivers, and process setting adjustment terms in AM (the SLM)

The vector $\mathbf{A} = \{a_k\} = \{\Delta\text{Setting}_k\}$ $(k = 1, 2, \ldots, 5)$ is used to represent the five AM process setting adjustment terms in Fig. 2, where the notations $a_1, a_2, \ldots, a_5$ are placed beside their corresponding terms. All a_k in $\mathbf{A}$ are normalized real numbers, i.e., $|a_k| \in [0, 1]$, while $a_k = 1$ indicates that the largest possible adjustment is made in the kth AM process setting when a PcV is derived from the PcP, and $a_k = 0$ indicates that zero adjustment is made. The detailed definition of each AM process setting adjustment term a_k is presented as follows:

(1) Support area adjustment a_1
Platform module design variations may require support area adjustments. For example, when an overhang portion is enlarged in a particular platform variant, the support area also needs to be increased accordingly and vice versa. Thus, a_1 can be calculated as

$$a_1 = \frac{\text{SA}_{\text{PcV}} - \text{SA}_{\text{PcP}}}{\text{SA}_{\text{PcVmax}} - \text{SA}_{\text{PcP}}} \tag{4}$$

where SA_{PcP} and SA_{PcV} represent support areas of the PcP and a derived PcV, respectively. The maximum possible support area adjustment value $(\text{SA}_{\text{PcVmax}} - \text{SA}_{\text{PcP}})$ is obtained when the largest projection area of the PcV on the build substrate, i.e., $\text{SA}_{\text{PcVmax}}$, is completely supported during the AM fabrication process.

(2) Orientation adjustment a_2
Platform module design variations may require adjustments in build orientation, in order to reduce support materials and improve surface smoothness [8]. Rotations about the Z axis (i.e., the axis parallel to the build direction) will not affect the build time and quality. Therefore, the orientation adjustment term a_2 is a function of rotations $\Delta\beta_X$ and $\Delta\beta_Y$ about the X and Y axes, respectively, expressed as

$$a_2 = \sqrt{\frac{1}{2}\left(\frac{\Delta\beta_X^{(\text{PcV}-\text{PcP})}}{\pi}\right)^2 + \frac{1}{2}\left(\frac{\Delta\beta_Y^{(\text{PcV}-\text{PcP})}}{\pi}\right)^2} \tag{5}$$

where the $(\text{PcV} - \text{PcP})$ superscript indicates that the orientation adjustment is made from the PcP to a PcV. When both $\Delta\beta_X$ and $\Delta\beta_Y$ take the largest possible orientation adjustment value π, a_2 has its maximum value of 1.

(3) Scan speed adjustment a_3
Scan speed can be adjusted to obtain the required surface quality and material properties of a particular platform variant [28]. The scan speed adjustment term a_3 is expressed as

$$a_3 = \Delta V/|\Delta V|_{\max} = (V_{\text{PcV}} - V_{\text{PcP}})/|\Delta V|_{\max} \tag{6}$$

where V_{PcP} and V_{PcV} are the scan speed values of the PcP and a derived PcV, respectively. $|\Delta V|_{\max}$ is the maximum possible adjustment range, beyond which the as-built material may have inferior properties due to insufficient energy density.

(4) Layer thickness adjustment a_4
Similar to the scan speed, layer thickness in AM can also be adjusted to obtain desired part qualities of a platform variant. The layer thickness adjustment a_4 is expressed as

$$a_4 = \Delta T/|\Delta T|_{\max} = (T_{\text{PcV}} - T_{\text{PcP}})/|\Delta T|_{\max} \tag{7}$$

where T_{PcP} and T_{PcV} are the layer thickness values of the PcP and a derived PcV, respectively. $|\Delta T|_{\max}$ is the maximum possible adjustment range, beyond which the build process may fail due to excessively thin or thick layers.

(5) Hatch density adjustment a_5
Hatch density can be adjusted to create different weights and strengths of platform variant modules. The hatch density adjustment a_5 is expressed as

$$a_5 = \Delta H = H_{\text{PcV}} - H_{\text{PcP}} \tag{8}$$

where H_{PcP} and H_{PcV} are the hatch density values of the PcP and a derived PcV, respectively. Since both H_{PcP} and H_{PcV} range from 0% to 100%, the $|a_5|$ value is in the scale of [0, 1].

Step B. Obtain the fuzzy total capacity cost in each activity
For each activity in AM production, the total capacity cost per unit part within a period (e.g., 1 month) is recorded. A cost value usually varies over time (for instance, February's value being different from January's value) due to unpredictable fluctuations in company operations. Therefore, the total capacity cost in activity i can be modeled by a triangular fuzzy number $C_i = [C_{iS}, C_{iM}, C_{iL}]$ [26], as formulated in Eq. (9), where the unit is dollar The smallest value C_{iS}, the most probable value C_{iM}, and the largest value C_{iL} are obtained from the company's historical data. As indicated in Fig. 1, the total capacity cost C_i is a basic term required to calculate the capacity time rate R_i

$$C_i = [C_{iS}, C_{iM}, C_{iL}]$$
$$\text{where } i = 1 \text{ and } 2 \text{ for activities 1 and 2, respectively} \tag{9}$$

Step C. Obtain the fuzzy practical capacity time in each activity
The practical capacity time per unit part in activity i can also be modeled by a triangular fuzzy number $T_i = [T_{iS}, T_{iM}, T_{iL}]$ obtained from historical data, as formulated in Eq. (10) where the unit is hour. As indicated in Fig. 1, the practical capacity time T_i is another basic term required to calculate the capacity time rate R_i

$$T_i = [T_{iS}, T_{iM}, T_{iL}]$$
$$\text{where } i = 1 \text{ and } 2 \text{ for activities 1 and 2, respectively} \tag{10}$$

If a company is starting out a new AM production line, AM fabrication test runs need to be conducted to initialize $[C_{iS}, C_{iM}, C_{iL}]$ and $[T_{iS}, T_{iM}, T_{iL}]$ values. When more data are accumulated in the company's operation, $[C_{iS}, C_{iM}, C_{iL}]$ and $[T_{iS}, T_{iM}, T_{iL}]$ can be

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnljTWeENb774MKM7HJ-nZUW6TxtlJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbebQw by Universita Di Pisa user on 25 November

further updated. The data accumulation is a common process in corporate costing approaches [25].

Step D. Calculate the fuzzy capacity cost rate

The fuzzy capacity cost rate R_i in activity i is the ratio of the total capacity cost C_i to the practical capacity time T_i, as formulated in the following equation:

$$R_i = \mu(C_i/T_i) \quad \text{where } i = 1 \text{ and } 2 \text{ for activities 1 and 2, respectively} \tag{11}$$

where the operator $\mu(A \otimes B)$ represents an α-cut fuzzy arithmetic ($\otimes$ can be $+, -, \times,$ or $/$) between two fuzzy numbers A and B.

The fuzzy capacity cost rate R_i is considered as the "unit cost" that can be used to calculate the total cost by multiplying it with time consumptions in each activity. The formulation of time equations is introduced in step E.

Step E. Formulate the fuzzy time equation

In activity i, the time consumption $t_{i\text{PcP}}$ of the PcP is expressed in a triangular fuzzy number to model the uncertainty, as formulated in Eq. (12). While $t_{i\text{PcP}}$ can be obtained from historical data of the reference platform design, the time consumption $t_{i\text{PcV}}$ of a particular PcV is a function of process setting adjustments. Mathematically, the time equation of a PcV in activity i is formulated in Eq. (13)

$$t_{i\text{PcP}} = \tau_i = [\tau_{iS}, \tau_{iM}, \tau_{iL}] \tag{12}$$

$$t_{i\text{PcV}} = \tau_i \prod_{j}^{N} [1 + \theta(\boldsymbol{a}^j)_i] = \tau_i \Theta_i = [\tau_{iS}\Theta_i, \tau_{iM}\Theta_i, \tau_{iL}\Theta_i] \quad \text{where } i = 1 \text{ and } 2 \text{ for activities 1 and 2, respectively} \tag{13}$$

where $\boldsymbol{a}^j$ is a set of relevant process setting adjustments that are taken as variables in activity i, represented as

$$\boldsymbol{a}^j \subseteq \mathbf{A} \tag{14}$$

where $\boldsymbol{A} = \{a_1, a_2, a_3, a_4, a_5\}$ are process setting adjustment terms defined in step A.

In Eq. (13), $\theta(\boldsymbol{a}^j)_i$ is the percentage increment of time consumption caused by an adjustment $\boldsymbol{a}^j$ in a process setting variant. The method of calculating $\theta(\boldsymbol{a}^j)_i$ values will be discussed separately in Sec. 4.3.

Step F. Calculate the activity cost

The activity cost AC_i in activity i is the α-cut fuzzy product of the activity time consumption and the capacity cost rate, as expressed in the following equation:

$$\begin{aligned} &\text{AC}_{i\text{PcP}} = \mu(t_{i\text{PcP}} \times R_i), \quad \text{for process setting platform (PcP)} \\ &\text{AC}_{i\text{PcV}} = \mu(t_{i\text{PcV}} \times R_i), \quad \text{for process setting variant (PcV)} \\ &\text{where } i = 1 \text{ and } 2 \text{ for activities 1 and 2, respectively} \end{aligned} \tag{15}$$

where both $\text{AC}_{i\text{PcP}}$ and $\text{AC}_{i\text{PcV}}$ are fuzzy numbers.

Step G. Calculate and defuzzify the total cost

As shown in Fig. 1, the last step of the FTDABC method is to calculate the fuzzy total cost TC_f in AM production, which is the α-cut fuzzy summation of activity cost values in activities 1 and 2, as formulated in Eq. (16). The fuzzy total cost $\text{TC}_f = \text{TC}_f(r)$ is a fuzzy membership function, while each value of the positive real variable $r \in R^+$ has a corresponding membership degree determined by $\text{TC}_f(r)$. $\text{TC}_f(r)$ is defuzzified using the centroid method to obtain the most probable crisp value of the total cost TC [29], as formulated in Eq. (17)

$$\begin{aligned} &\text{TC}_{f\text{PcP}} = \mu(\text{AC}_{1\text{PcP}} + \text{AC}_{2\text{PcP}}), \quad \text{for process setting platform (PcP)} \\ &\text{TC}_{f\text{PcV}} = \mu(\text{AC}_{1\text{PcV}} + \text{AC}_{2\text{PcV}}), \quad \text{for process setting variant (PcV)} \end{aligned} \tag{16}$$

$$\begin{aligned} \text{TC}_{\text{PcP}} &= \frac{\int_{r_{\min}}^{r_{\max}} \text{TC}_{f\text{PcP}}(r) r dr}{\int_{r_{\min}}^{r_{\max}} \text{TC}_{f\text{PcP}}(r) dr}, \quad \text{for process setting platform (PcP)} \\ \text{TC}_{\text{PcV}} &= \frac{\int_{r_{\min}}^{r_{\max}} \text{TC}_{f\text{PcV}}(r) r dr}{\int_{r_{\min}}^{r_{\max}} \text{TC}_{f\text{PcV}}(r) dr}, \quad \text{for process setting variant (PcV)} \end{aligned} \tag{17}$$

4.3 Time Equation Formulation Using ANFIS. In this section, the time equation formulation method in step E is explained in detail. An ANFIS-based expert system [30] is used to predict each activity's time consumption as a function of process setting adjustments. There are several benefits of implementing ANFIS: (1) time consumptions can be estimated quickly, without the necessity of importing exact computer-aided design (CAD) models into computer-aided manufacturing (CAM) software [9,31], and hence, a large number of process setting adjustment values can be evaluated in the later stage of FS's boundary searching; (2) the company's historical data can be utilized to create an empirical model, and new empirical data generated in the future can be used to further update the existing ANFIS; and (3) uncertainties are captured in time equations through the use of fuzzy membership functions.

An ANFIS is created to map each $\boldsymbol{a}^j$ to $\theta(\boldsymbol{a}^j)_i$ in time equations, and hence, each activity in the FTDABC method may contain multiple ANFIS. A set of "IF (antecedent)–THEN (consequence)" rules established from prior data are used to initialize a first-order Takagi–Sugeno (TS) fuzzy model [32]. A sample set of rules is shown in the following equation for illustration:

$$\begin{aligned} &\text{Input variable}: \ \boldsymbol{a}^j = \{a_1, a_2\} \\ &R_1: \ \text{IF } (a_1 \text{ is } \lambda_{11} \text{ AND } a_2 \text{ is } \lambda_{21}) \\ &\quad \text{THEN } \theta(\boldsymbol{a}^j)_i' = p_{11}a_1 + p_{12}a_2 + q_1 \\ &R_2: \ \text{IF } (a_1 \text{ is } \lambda_{12} \text{ AND } a_2 \text{ is } \lambda_{22}) \\ &\quad \text{THEN } \theta(\boldsymbol{a}^j)_i'' = p_{21}a_1 + p_{22}a_2 + q_2 \end{aligned} \tag{18}$$

In Eq. (18), λ_{kl} is a linguistic term (e.g., "high," "low," etc.) defined by an antecedent fuzzy membership function (Gaussian), as formulated in Eq. (19). The consequent is defined by a linear function $\boldsymbol{p}_l^T \boldsymbol{a}^j + q_l$. The model output $\theta(\boldsymbol{a}^j)_i$ is the weighted average of consequent function values in all rules, as formulated in Eqs. (20) and (21)

$$\lambda_{kl}(a_k, c_{kl}, \sigma_{kl}) = \exp\left(-\frac{(a_k - c_{kl})^2}{2\sigma_{kl}{}^2}\right) \tag{19}$$

$$\theta(\boldsymbol{a}^j)_i = \sum_{l=1}^{L} \varphi_l (\boldsymbol{p}_l^T \boldsymbol{a}^j + q_l) \tag{20}$$

$$\varphi_l = \frac{\prod_{k=1}^{K} \lambda_{kl}}{\sum_{l=1}^{L} \left(\prod_{k=1}^{K} \lambda_{kl} \right)} \tag{21}$$

where $(L = 2)$ is the total number of rules and $(K = 2)$ is the number of input variables (i.e., process setting adjustment terms $\{a_1, a_2\}$). In the above equations, expert knowledge is applied in: (1) selecting relevant AM process setting adjustment terms that are input to the TS model; (2) partitioning the input domain and determining the number of fuzzy membership functions; and (3)

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnljTWeENb774MKM7HJ-nZUW6TxtlJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeBQw by Universita Di Pisa user on 25 November

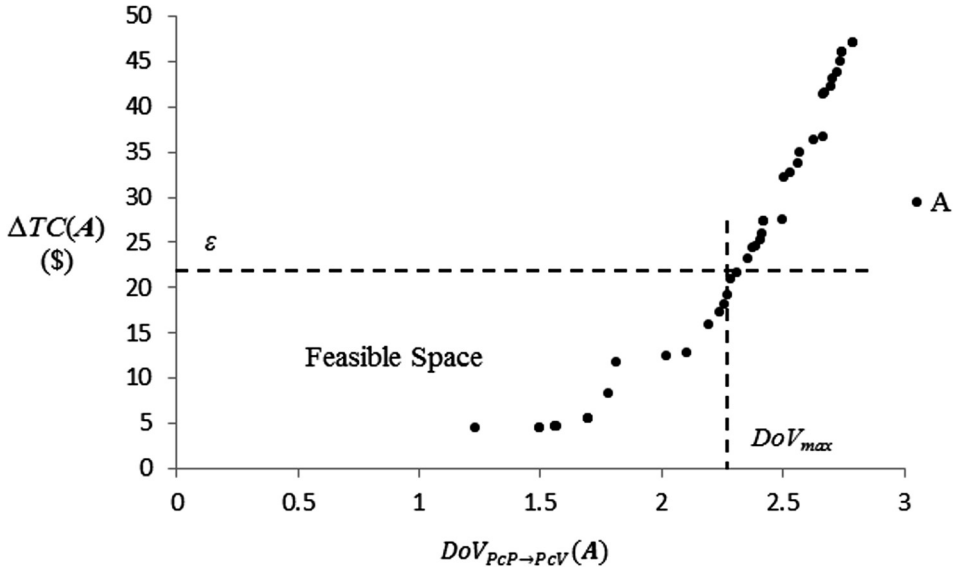


Fig. 3 A sample AM process setting adjustments' FS

initializing the fuzzy rules with guessed parameters $(\boldsymbol{p}_l^T, q_l, c_{kl}, \sigma_{kl})$ that are to be refined later in the training process.

Historical data pairs $(\mathbf{a}^j, \theta(\mathbf{a}^j)_i)$ are used to train the TS model by optimizing the parameters $(\boldsymbol{p}_l^T, q_l, c_{kl}, \sigma_{kl})$. A hybrid learning algorithm (a combination of least-squares estimates and gradient-based optimization) is used to update the parameters [30].

Existing AM costing software, including online and offline applications [33], requires the input of a part's digital model in the format of stereolithography (STL) or CAD files. However, the existing file-based costing algorithms lack the capability of predicting cost changes as an effect of relative design adjustments, unless new CAD/STL files with updated design variables are imported to the costing software. In contrast, the proposed FTDABC-based costing algorithm is able to automatically estimate the effect of part design adjustments on cost changes in multiple cost drivers, without the necessity of updating and importing new STL/CAD files. This capability allows the proposed costing algorithm to evaluate a large number of part design parameters that are repeatedly updated for hundreds of iterations in the optimization process as discussed in Sec. 4.4.

4.4 AM Process Setting Adjustments' FS Boundary Searching. As indicated in Fig. 1, after the proposed costing model is established, the next step is to identify the FS boundary of process setting adjustment values in order to constraint cost increments. The boundary searching is performed by solving a multi-objective optimization problem, whose input variable is the vector **A** containing five AM process adjustment terms defined in Sec. 4.2.

As shown in Eq. (22), cost increments (defined in Eqs. (1)–(3)) of deriving a PcV from the PcP can be computed as the difference between $\mathrm{TC}_{\mathrm{PcV}}$ and $\mathrm{TC}_{\mathrm{PcP}}$, which are both calculated by Eq. (17) in the FTDABC method

$$\mathrm{CostInc} = \Delta\mathrm{TC}(\mathbf{A}) = \mathrm{TC}_{\mathrm{PcV}}(\mathbf{A}) - \mathrm{TC}_{\mathrm{PcP}} \tag{22}$$

The metric named "degree of variation (DoV)" is defined to measure the difference between a PcV and the PcP. The DoV is formulated as the Euclidean norm of the vector **A**

$$\begin{aligned} &\mathrm{DoV}_{\mathrm{PcP}\to\mathrm{PcV}}(\mathbf{A}) \;:= \|\mathbf{A}\|^2 = \sum_k a_k^2, \text{ where } k = 1, 2, \ldots, 5 \\ &|a_k| \in [0, 1] \\ &\mathrm{DoV}_{\mathrm{PcP}\to\mathrm{PcV}}(\mathbf{A}) \in [0, 5] \end{aligned} \tag{23}$$

Since the research objective is to reduce cost increments caused by AM process setting adjustments, the $\Delta\mathrm{TC}(\mathbf{A})$ value should be minimized in the multi-objective optimization problem for FS boundary searching. However, the DoV value implying the size of the process setting's FS should be maximized in order to provide more freedoms in manipulating the AM process. The complete multi-objective optimization problem can be formulated as follows:

$$\begin{aligned} &\text{Variable: } \mathbf{A} = \{a_k\} \\ &\text{Objective functions: } \begin{cases} \min\limits_{\mathbf{A}} & \Delta\mathrm{TC}(\mathbf{A}) \\ \max\limits_{\mathbf{A}} & \mathrm{DoV}_{\mathrm{PcP}\to\mathrm{PcV}}(\mathbf{A}) \end{cases} \\ &\text{Bound constraint: } |a_k| \in [0, 1] \end{aligned} \tag{24}$$

A multi-objective genetic algorithm (GA) [34] is used to solve the optimization problem. As the result, Pareto set of **A** are identified, as illustrated in Fig. 3, where each point on the plot represents an instance of **A**. A predetermined cost increment upper limit ε is used to contain the process setting adjustments' FS. The Pareto front below the horizontal dashed line $(\Delta\mathrm{TC}(\mathbf{A}) = \varepsilon)$ defines the FS boundary. The ε value also defines the maximum allowable DoV ($\mathrm{DoV}_{\max} = 2.3$ in Fig. 3). Any process setting adjustment that has a DoV value larger than the $\mathrm{DoV}_{\max}$ is infeasible due to cost increments larger than the upper limit.

4.5 Mapping AM Process Settings to Variable Product Platform Design Constraints. In this section, we present a mathematical method for mapping AM process settings to variable platform design constraints.

The *j*th platform variant design in the *i*th product variant $(\boldsymbol{X}_{\mathrm{PV},ij})$ is expressed as a vector of n design parameters, each of which is a scalar variable x_p^k. $\boldsymbol{X}_{\mathrm{PV},ij}$ is the product of the platform design parameter variation matrix $\boldsymbol{D}_{ij}$ and the reference platform design parameter vector $\boldsymbol{X}_{R,j}$, as formulated in Eq. (25). $\boldsymbol{D}_{ij}$ is an $n \times n$ diagonal matrix with its diagonal elements d_{kk} being dimensionless positive real numbers. $\boldsymbol{X}_{R,j}$ represents an initial or currently existing design of the variable platform module, and it is the reference based on which all platform variant designs $\boldsymbol{X}_{\mathrm{PV},ij}$ are derived. Diagonal elements d_{kk} in $\boldsymbol{D}_{ij}$ control the variation of design parameters in $\boldsymbol{X}_{\mathrm{PV},ij}$ across different product variants (totally N) in the family

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnIj7WeENb774MKM7HJ-nZUW6TxtlJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeBQw by Universita Di Pisa user on 25 November

$$\boldsymbol{X}_{\mathrm{PV},ij} = \boldsymbol{D}_{ij}\boldsymbol{X}_{R,j} \Rightarrow \begin{Bmatrix} x_p^1 \\ x_p^2 \\ \vdots \\ x_p^n \end{Bmatrix}_j = \begin{pmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{nn} \end{pmatrix}_j \begin{Bmatrix} x^1 \\ x^2 \\ \vdots \\ x^n \end{Bmatrix}_j$$

where $i = 1, 2, \ldots, N$ (25)

Platform variants' design parameter variations are subject to constraints imposed by process setting adjustments' FS, which is determined by the maximum acceptable AM cost increments, as discussed in Sec. 4.4. The constraints are expressed in an inequality equation as shown below:

$$\boldsymbol{D}_{ij}^{\min} \le \boldsymbol{D}_{ij} \le \boldsymbol{D}_{ij}^{\max} \tag{26}$$

Although relationships between AM process settings and some geometric design constraints were experimentally studied in previous research [35–38], predictive mathematical models have not been established. In practice, design rules in AM process are formed mainly empirically based on designers' experience and knowledge. Therefore, in this research, a knowledge-based expert system is used to establish the mapping from the process setting adjustments' FS boundary to product platform design constraints. A Mamdani-type fuzzy inference system, or a Mamdani model [32], is applied to create the expert system. A point $\mathbf{A}^* = \{a_k\}^*$ on the process setting adjustments' FS boundary is chosen as the input to the Mamdani model, while $\boldsymbol{D}_{ij}^{\min}$ and $\boldsymbol{D}_{ij}^{\max}$ are the outputs. The selection of $\mathbf{A}^*$ depends on the desired DoV and acceptable cost increment values.

IF–THEN design rules with linguistic antecedent and consequent terms expressed in fuzzy membership functions are formulated in the Mamdani model to represent expert knowledge [32]. As an example, design parameters and constraints of a honeycomb feature fabricated by the SLM process are expressed in Eq. (27), and an example design rule R is formulated in Eq. (28). As defined in Sec. 4.2, a_1, a_2, a_3, a_4, and a_5 (in $\mathbf{A}$) represent support area adjustment, orientation adjustment, scan speed adjustment, layer thickness adjustment, and hatch density adjustment in the SLM, respectively

$$\begin{aligned}
\boldsymbol{X}_{R,j} &= \begin{Bmatrix} x^1 \\ x^2 \\ x^3 \\ x^4 \end{Bmatrix} = \begin{Bmatrix} \text{bounding box length} \\ \text{bounding box height} \\ \text{cell size} \\ \text{wall thickness} \end{Bmatrix} = \begin{Bmatrix} B \\ H \\ C \\ T \end{Bmatrix} \\
\boldsymbol{D}_{ij} &= \begin{pmatrix} d_B & 0 & 0 & 0 \\ 0 & d_H & 0 & 0 \\ 0 & 0 & d_C & 0 \\ 0 & 0 & 0 & d_T \end{pmatrix} \\
\boldsymbol{D}_{ij}^{\min} &= \begin{pmatrix} d_B^{\min} & 0 & 0 & 0 \\ 0 & d_H^{\min} & 0 & 0 \\ 0 & 0 & d_C^{\min} & 0 \\ 0 & 0 & 0 & d_T^{\min} \end{pmatrix} \\
\boldsymbol{D}_{ij}^{\max} &= \begin{pmatrix} d_B^{\max} & 0 & 0 & 0 \\ 0 & d_H^{\max} & 0 & 0 \\ 0 & 0 & d_C^{\max} & 0 \\ 0 & 0 & 0 & d_T^{\max} \end{pmatrix}
\end{aligned} \tag{27}$$

Input variable: $\mathbf{A} = \{a_k\} = \{\Delta\text{Setting}\}$

where $k = 1, 2, \ldots, 5$ for five process setting adjustment terms

R: IF (a_1 is high)

$$\text{THEN} \begin{cases} d_L^{\min} \text{ is small; } d_L^{\max} \text{ is large} \\ d_H^{\min} \text{ is small; } d_H^{\max} \text{ is large} \\ d_C^{\min} \text{ is small; } d_C^{\max} \text{ is large} \\ d_T^{\min} \text{ is small; } d_T^{\max} \text{ is large} \end{cases} \tag{28}$$

In the above example, R is formulated based on the knowledge that if the support area of the variable platforms (i.e., the honeycomb feature) has a large allowable adjustment range (i.e., a_1 is high), the honeycomb feature's overall dimensions and cell size can be increased, which results in larger overhang areas. The mathematic principles of rule-based fuzzy inference in a Mamdani model can be found in literatures [32]; hence, they are not elaborated in this paper. The Mamdani model is implemented in MATLAB®.

4.6 Variable Platform Design Optimization in a Product Family. After design constraints of variable platform modules are identified in terms of $\boldsymbol{D}_{ij}^{\min}$ and $\boldsymbol{D}_{ij}^{\max}$, design parameters of each platform variant in a product family are optimized to maximize the customer perceived utility. A part-worth utility function [39] of the ith product variant is formulated as

$$U_i(\boldsymbol{D}_{ij}) = \sum_{\varphi=1}^{\Phi} \beta_{p_\varphi}[p_\varphi(\boldsymbol{D}_{ij}, \boldsymbol{X}_{R,j}, \boldsymbol{X}_U)] \tag{29}$$

where $\boldsymbol{X}_U$ represents unique modules of the ith product that are not shared by others in the family; p_φ is the φth performance indictor of the product determined by both platform modules and unique modules; and β_{p_φ} is the utility weight assigned to each p_φ. The customer perceived utility is a measure of product performance in terms of customer preference in a particular market segment [39].

The platform variant design optimization problem in a particular market segment is formulated as follows:

$$\begin{aligned}
&\text{Variable: } \boldsymbol{D}_{i,j} \\
&\text{Objective function: } \max U_i(\boldsymbol{D}_{ij}) \\
&\text{Bound constraint: } \boldsymbol{D}_{ij}^{\min} \le \boldsymbol{D}_{ij} \le \mathbf{D}_{ij}^{\max}
\end{aligned} \tag{30}$$

5 Case Studies

A family of TRAXXAS R/C racing cars [40] is used to demonstrate usefulness of the proposed cost-driven variable platform design methodology. In case study I, the design of a single variable platform module is analyzed to explore the AM process setting adjustments' FS and design parameter constraints. In case study II, the R/C racing car family containing multiple additive manufactured variable platform modules is optimized to maximum customer perceived utility in each market segment.

5.1 Case Study I: Single Variable Platform Module Design. The car bumper is chosen as a variable platform module in the R/C car family. The result from AM process setting adjustments' FS exploration and design constraints identification is presented in this section.

The conceptual honeycomb-based bumper design is shown in Fig. 4. Honeycomb and lattice structures have found broad applications in automotive and aerospace due to its high

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnlj7WeENb774MKM7HJ-nZUW6TxtlJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeBQw by Universita Di Pisa user on 25 November

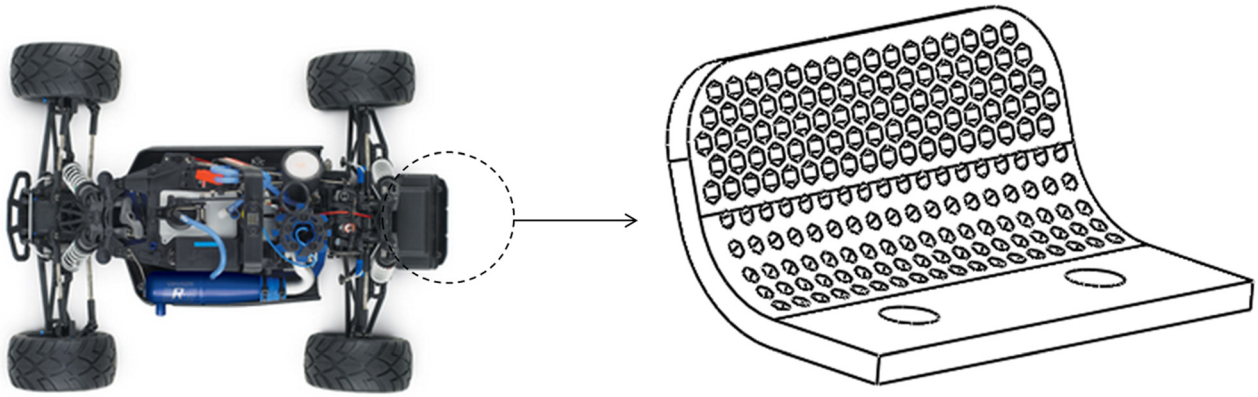


Fig. 4 Bumper design with the honeycomb feature

Table 1 Reference design parameters of the bumper

k	Key design parameters	$X_{R\ (\mathrm{mm})}$
1	Bounding box length (B)	55
2	Bounding box height (H)	20
3	Cell size (C)	3
4	Wall thickness (T)	2

Table 2 The SLM process settings in the PcP of the reference bumper design

Support area (mm^2)	Orientation (rad)	Scan speed (mm/s)	Layer thickness (mm)	Hatch density (%)
210	0	550	0.05	80

strength-to-weight ratio [41,42]. Key design parameters ($\mathbf{X}_{R,j} = [B\ H\ C\ T]^T$) of the honeycomb-based bumper have been defined in Eq. (27) in Sec. 4.5. The reference bumper design parameters are listed in Table 1, based on which each platform variant in the racing car family can be optimized in its respective market segment.

The SLM process setting of the reference bumper design is treated as the PcP, with the values listed in Table 2.

The application of the FTDABC method for AM cost estimation is illustrated in Sec. 5.1.1. The resultant AM process setting adjustments' FS boundary is presented in Sec. 5.1.2. The variable platform design constraints are calculated in Sec. 5.1.3.

5.1.1 Cost Estimation by the FTDABC Method. At the data preparation stage (activity 1), SOLIDWORKS® was used to create and modify digital models; MAGICS[2] was used to plan build orientations and support structures; and AUTOFAB (see footnote no. 2) was the CAM software used to set SLM process parameters and to generate ready-to-print machine codes. During the build process (activity 2), SLM® 250HL manufactured by SLM Solutions GmbH[3] (Lubeck, Germany) was used to produce components in aluminum alloys. The build chamber was filled with argon gas. Engineers monitored the build process when the machine ran from the start to the end.

The ANFIS model was used to formulate time equations in the FTDABC method. For brevity, the ANFIS model of only activity 1 is shown here for illustration. The first-order ANFIS output functions $\theta(\boldsymbol{a}^j)_i$ in activity 1 are written in Eq. (31), with their trained coefficients and corresponding rules listed in Table 3

[2]http://software.materialise.com
[3]http://www.stage.slm-solutions.com

Table 3 First-order output functions' coefficients in corresponding rules

	Output function (THEN)				
	$\theta(a^1)_1$			$\theta(a^2)_1$	
Antecedent (IF)	p'	p''	q'	p''	q''
a_1 is low AND a_2 is low	2.570	0.986	0.070		
a_1 is medium OR a_2 is medium	1.139	2.041	−0.296		
a_1 is high OR a_2 is high	0.956	1.295	−0.503		
a_5 is high				0.664	0.002
a_5 is medium				0.587	−0.157
a_5 is high				1.365	−0.865

Table 4 Fuzzy time consumption $\tau_i = [\tau_{iS}, \tau_{iM}, \tau_{iL}]$

Activity (i)	τ_{iS} (hr)	τ_{iM} (hr)	τ_{iL} (hr)
Data preparation ($i=1$)	0.6	1.0	1.2
Build process ($i=2$)	4.0	4.6	5.5

Table 5 The total capacity cost and practical capacity rate in the SLM production

	C_i (\$/unit)			T_i (hr)		
Activity (i)	C_{iS}	C_{iM}	C_{iL}	T_{iS}	T_{iM}	T_{iL}
Data preparation ($i=1$)	50	80	90	0.6	1.4	1.9
Build process ($i=2$)	95	150	210	2	5	11

$$\theta(\boldsymbol{a}^1)_1 = \theta(\{a_1, a_2\})_1 = p'a_1 + p''a_2 + q'$$
$$\theta(\boldsymbol{a}^2)_1 = \theta(\{a_5\})_1 = p'''a_5 + q''$$
$$\text{where} \begin{cases} a_1 = \{\Delta\text{support area}\} \\ a_2 = \{\Delta\text{orientation}\} \\ a_5 = \{\Delta\text{hatch density}\} \end{cases} \quad (31)$$

The fuzzy time consumptions τ_i in activities 1 and 2 of producing the reference bumper design are listed in Table 4. With τ_i and $\theta(\boldsymbol{a}^j)_i$ obtained, time equations $t_{i\mathrm{PcV}} = \tau_i \prod_j^N [1 + \theta(\boldsymbol{a}^j)_i]$ can be calculated.

The total capacity costs C_i and practical capacity rates T_i in activities 1 and 2 are listed in Table 5. The fuzzy membership functions of resultant capacity cost rates $R_i = \mu(C_i/T_i)$ are plotted

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnlj7WeENb774MKM7HJ-nZUW6TxtlJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeQw by Universita Di Pisa user on 25 November

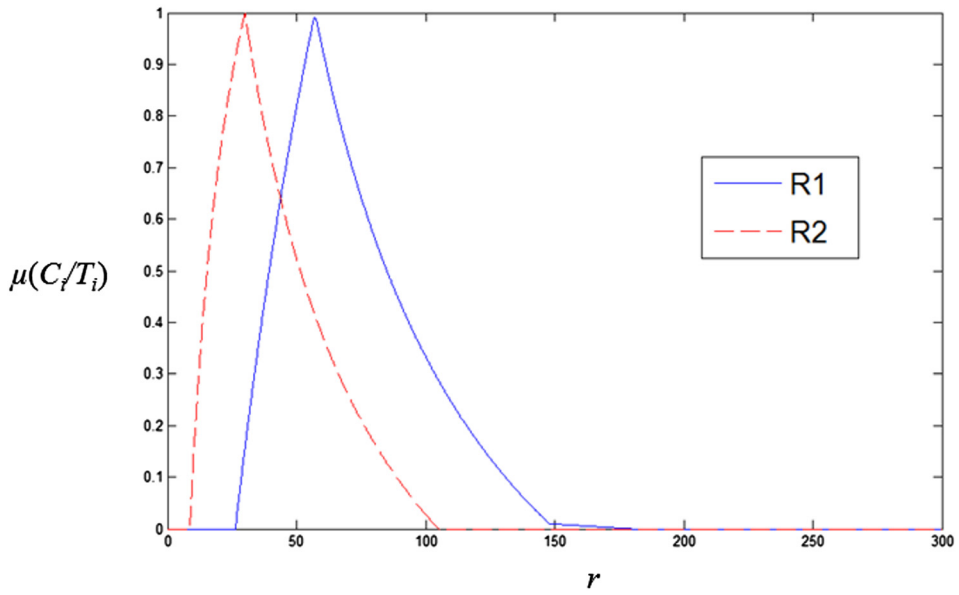


Fig. 5 Calculated fuzzy capacity cost rate $R_i = \mu(C_i/T_i)$

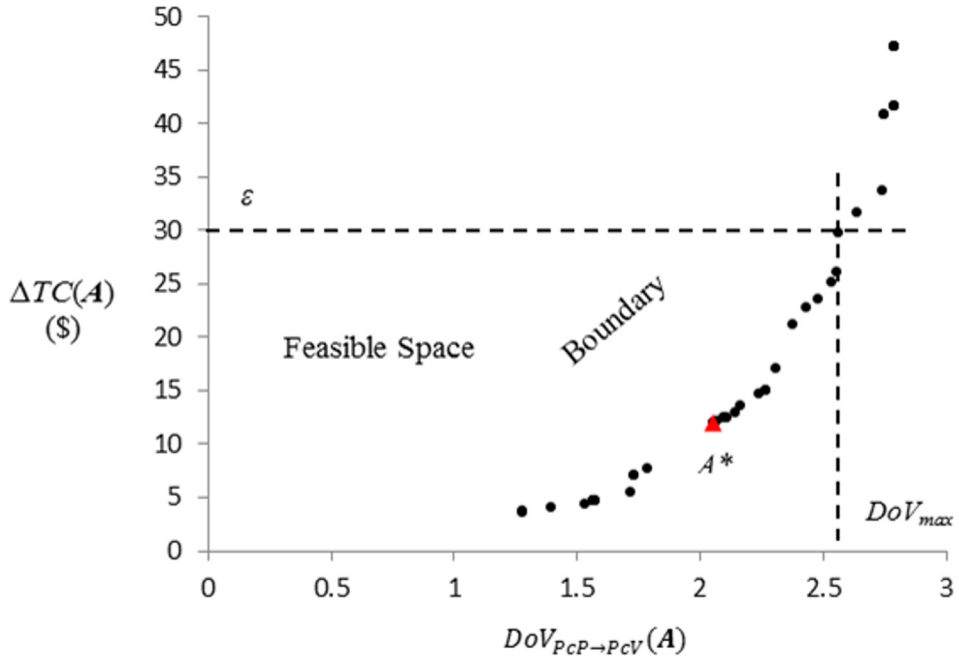


Fig. 6 Identified FS with the boundary

in Fig. 5, where the function variable r is shown on the horizontal axis, and the membership degree determined by $R_i = \mu(C_i/T_i)$ is shown on the vertical axis. The cost of each activity AC_i and the resultant total cost TC can be calculated from fuzzy capacity cost rate R_i and time equations $t_{i\text{PcV}}$ using Eqs. (15)–(17).

5.1.2 Process Setting Adjustments' FS Boundary Searching. Figure 6 shows the Pareto front (computed by multi-objective GA) that consists of 50 Pareto-optimal solutions indicated by the dots. The cost increment upper limit (ε) was set at \$30 per unit part. The Pareto front bound by the horizontal dashed line $\varepsilon = 30$ forms the FS boundary of AM process setting adjustments. The maximum possible degree of variation ($\text{DoV}_{\max}$) is 2.57.

5.1.3 Honeycomb Structure Design Parameter Constraints. Assuming that the company wanted to limit the cost increments of producing each bumper variant at \$13 per unit, the point $\mathbf{A}^*$ (the triangular marker in Fig. 6) was selected on the process setting adjustment's FS boundary, so that the maximum acceptable cost increment is $\Delta\text{TC} = \$12.41 < \$13$ and the maximum possible DoV is 2.09. Equation (32) shows the values in vector $\mathbf{A}^*$, indicating the maximum allowable adjustments on the PcP when deriving a PcV. For example, the first element in $\mathbf{A}^*$ is $a_1^* = 0.088$, indicating that the support area in the PcP of any bumper variant can be increased by no more than 0.088 (or 8.8%) of the largest possible support area adjustment value (1990 mm^2). If the support area is increased beyond the allowable limit, extra cost larger than the maximum ΔTC value will be induced

$$\mathbf{A}^* = \{a_k^*\} = \{0.088, 0.870, -0.421, -0.818, 0.2\} \quad (32)$$

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5fTPnlj7WeENb774MKM7HJ-nZUW6TxtlJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbebQw by Universita Di Pisa user on 25 November

Using the fuzzy rules for the honeycomb structure design mentioned in Sec. 4.5, the constraints of design parameters ($\boldsymbol{D}_{ij}^{\min}$ and $\boldsymbol{D}_{ij}^{\max}$) were calculated by the Mamdani model using $\mathbf{A}^{*}$ as the input. The results are shown in Eqs. (33) and (34)

$$\boldsymbol{D}_{ij}^{\min} = \begin{pmatrix} 0.54 & 0 & 0 & 0 \\ 0 & 0.65 & 0 & 0 \\ 0 & 0 & 0.52 & 0 \\ 0 & 0 & 0 & 0.58 \end{pmatrix}$$
$$\boldsymbol{D}_{ij}^{\max} = \begin{pmatrix} 1.52 & 0 & 0 & 0 \\ 0 & 1.55 & 0 & 0 \\ 0 & 0 & 2.05 & 0 \\ 0 & 0 & 0 & 2.57 \end{pmatrix} \tag{33}$$

The diagonal elements of $\boldsymbol{D}_{ij}^{\min}$ and $\boldsymbol{D}_{ij}^{\max}$ calculated in Eq. (33) represent the minimum and maximum allowable scale factors applied to the reference design parameters ($\mathbf{X}_{R,j} = [B\ H\ C\ T]^T$). Therefore, the bound constraint of any bumper variant's design parameters can be explicitly expressed as

$$\begin{aligned} [B^{\min}\ H^{\min}\ C^{\min}\ T^{\min}]^T &= [0.54B\ 0.65H\ 0.52C\ 0.58T]^T \\ [B^{\max}\ H^{\max}\ C^{\max}\ T^{\max}]^T &= [1.52B\ 1.55H\ 2.05C\ 2.57T]^T \end{aligned} \tag{34}$$

Recalling the reference design parameter $\boldsymbol{X}_{R,j}$ listed in Table 1, the minimum and maximum bumper variant design parameters are calculated using Eq. (34) and the relationship $\boldsymbol{X}_{\mathrm{PV},ij} = \boldsymbol{D}_{ij}\boldsymbol{X}_{R,j}$ in Eq. (25). The results are shown in Table 6 and used as bound constraints in platform design optimization as illustrated in case study II in Sec. 5.2.

5.2 Case Study II: Multiple Platform Variants Design Optimization in Different Market Segments. A family of two R/C racing cars were designed for two different market segments: "short-course race truck" and "on-road sedan," as shown in Table 7. Short-course race trucks ran on rugged grounds with random presence of sands and rocks. While on-road sedans ran on flat, smooth, and relatively straight tracks. In this case study, multiple variable platform modules fabricated by AM were optimized to maximize customer perceived utility as a measure of product performance.

Table 7 also displays two types of unique modules, i.e., engine and suspension, in each product variant. For simplicity, it was assumed that these two unique modules are purchased off-the-shelf, and hence, their designs were assumed constant. Mathematically, the unique module design parameter $\boldsymbol{X}_U$ is a vector of integers as shown in the following equation, where E is 1 for type 1 engine ("nitro") and 2 for type 2 engine ("electric"); S is 1 for type 1 suspension ("rocker-actuated") and 2 for type 2 suspension ("stand-up"):

$$\mathbf{X}_U = \begin{Bmatrix} E \\ S \end{Bmatrix};\ E, S \in \{1, 2\} \tag{35}$$

Variable platform modules shared by both cars in the family included: (1) bumper, (2) chassis, and (3) wheel. Conceptual drawings are shown in Table 8. These three modules were chosen as variable platforms because: (1) the modules were fundamental components in cars; (2) the modules could have similar but nonidentical geometries in different product variants; and (3) the design parameters of the modules affected various aspects of car performance.

Values in the reference platform design parameter vector $\boldsymbol{X}_{R,j}$ of each variable platform module are listed in Table 9. Table 9 also contains the design parameter constraints ($\boldsymbol{D}_{ij}^{\min}$ and $\boldsymbol{D}_{ij}^{\max}$) calculated by the method illustrated in case study I.

Four performance indicators were used as factors in customer utility calculation: the weight-reduction indicator W, acceleration

Table 6 Minimum and maximum bumper design parameters in platform variants

Design parameter	B (mm)	H (mm)	C (mm)	T (mm)
Minimum	29.7	13.0	1.6	1.2
Maximum	83.6	31.0	6.2	5.1

Table 7 The family of R/C racing cars. Adapted from Ref. [40].

	Market segment 1 Short-course race truck	Market segment 2 On-road sedan
Overview		
Unique module: engine (off-the-shelf)	 (Type 1: nitro)	 (Type 2: electric)
Unique module: suspension (off-the-shelf)	 (Type 1: rocker-actuated)	 (Type 2: stand-up)

Table 8 Variable platform modules in the R/C racing car family

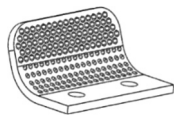
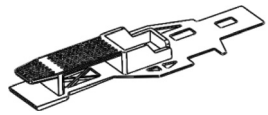
j	Variable platform module	Conceptual design drawing
1	Bumper	
2	Chassis	
3	Wheel	

Table 9 Key design parameters of variable platform modules

Platform module (j)	k	Key design parameters	$\boldsymbol{X}_{R,j}$	$d_{kk}^{\min}$	$d_{kk}^{\max}$
Bumper	1	Bounding box length (B)	55 mm	0.54	1.52
($j = 1$)	2	Bounding box height (H)	20 mm	0.65	1.55
	3	Cell size (C)	3 mm	0.52	2.05
	4	Wall thickness (T)	2 mm	0.58	2.57
Chassis	1	Overall length (L)	300 mm	0.86	1.22
($j = 2$)	2	Overall width (W)	100 mm	0.75	1.20
	3	Volume fraction (V)	70%	0.67	1.41
Wheel	1	Average cell size (R)	5 mm	0.56	2.30
($j = 3$)	2	Cell wall thickness (D)	2 mm	0.87	2.51

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnIj7WeENb774MKM7HJ-nZUW6TxtdJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeQw by Universita Di Pisa user on 25 November

Table 10 Optimal design parameters and resultant maximum customer utilities

	Bumper				Chassis			Wheel		Customer utility		
Market segment	d_B	d_H	d_C	d_T	d_L	d_W	d_M	d_R	d_D	$U_i(\boldsymbol{D}_{\text{optimal}})$	$U_i(\boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j})$	$\%\ (\Delta U)_i$
Variable platforms (short-course race truck)	0.54	0.65	2.01	0.63	0.89	0.75	0.67	0.75	0.99	1.78	1.61	10.6
Variable platforms (on-road sedan)	0.55	0.66	1.59	0.58	0.86	0.75	0.67	0.84	2.08	2.29	1.97	16.2
Consistent platforms ($\boldsymbol{D}_{1j} = \boldsymbol{D}_{2j}$)	0.55	0.67	2.03	0.81	0.86	0.78	0.70	0.88	2.45			

Table 11 Costs of the optimized variable platforms and consistent platforms

	AM production costs ($/unit)			$\%\ (\Delta C)_i$		
	Bumper	Chassis	Wheel	Bumper	Chassis	Wheel
$C_1(\boldsymbol{D}_{\text{optimal}})$ optimized variable platform (short-course race truck)	26	85	96	−25.7	−4.5	−14.3
$C_2(\boldsymbol{D}_{\text{optimal}})$ optimized variable platform (on-road sedan)	32	82	100	−8.6	−7.9	−10.7
$C(\boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j})$ optimized consistent platform	35	89	112			

indicator X, range indicator Y, and shock-resistance indicator Z. Part-worth utility functions are formulated in Eq. (36) for both short-course race trucks (U_1) and on-road sedans (U_2). Both U_1 and U_2 are objective functions that are maximized in variable platform design optimization

$$\begin{aligned} &\max: U_1(\boldsymbol{D}_{1j}) = 0.23W_1 + 0.27X_1 + 0.19Y_1 + 0.31Z_1 \\ &\max: U_2(\boldsymbol{D}_{2j}) = 0.29W_2 + 0.29X_2 + 0.26Y_2 + 0.16Z_2 \end{aligned} \tag{36}$$

GA implemented in MATLAB® was used to find optimal platform variant design parameters in each product variant in its corresponding market segment. The optimization result is shown in Table 10, where diagonal element values d of the optimal platform design parameter variation matrix $\boldsymbol{D}_{ij}$ and the corresponding maximum customer utility $U_i(\boldsymbol{D}_{\text{optimal}})$ are listed. Two product variants in the R/C racing car family were optimized independently in their corresponding market segments, and hence, the variable platform design parameters of the truck are different from those of the sedan, as shown in the first two rows of Table 10.

In order to compare the performance of variable product platforms with consistent platforms implemented in conventional product families, another design optimization problem was formulated with the objective function written as the linear combination of U_1 and U_2 (Eq. (37)). For simplicity, both weight values ω_1 and ω_2 were assumed 1

$$\begin{aligned} &\max: Y = \omega_1 U_1(\boldsymbol{D}_{1j}) + \omega_2 U_2(\boldsymbol{D}_{2j}) \\ &\text{where } \boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j} \end{aligned} \tag{37}$$

As indicated in Eq. (37), $\boldsymbol{D}_{1j}$ equals $\boldsymbol{D}_{2j}$ since design parameters of consistent platforms were shared by all product variants in the family. Optimal consistent platform design parameters are shown in the bottom row of Table 10, which also shows utility values $U_i(\boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j})$ when consistent platforms are applied. In Table 10, the positive values of the percentage difference $(\%(\Delta U)_i)$ between $U_i(\boldsymbol{D}_{\text{optimal}})$ and $U_i(\boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j})$ (Eq. (38)) indicate that the application of variable platforms has improved customer utility in both market segments

$$\%(\Delta U)_i = \frac{U_i(\boldsymbol{D}_{\text{optimal}}) - U_i(\boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j})}{U_i(\boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j})} \times 100\% \tag{38}$$

Table 11 shows the costs of three optimized variable platforms for short-course race truck $(C_1(\boldsymbol{D}_{\text{optimal}}))$ and on-road sedan $(C_2(\boldsymbol{D}_{\text{optimal}}))$, and the costs of optimized consistent platforms $(C(\boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j}))$. The percentage cost difference $\%(\Delta C)_i$ between an optimized variable platform and its consistent platform counterparts is calculated as

$$\%(\Delta C)_i = \frac{C_i(\boldsymbol{D}_{\text{optimal}}) - C(\boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j})}{C(\boldsymbol{D}_{1j} \equiv \boldsymbol{D}_{2j})} \times 100\% \tag{39}$$

The negative $\%(\Delta C)_i$ values shown in Table 11 indicate that the optimized variable platforms have lower cost than conventional consistent platforms. Therefore, combining the results shown in Tables 10 and 11, it can be demonstrated that the proposed cost-driven design methodology for variable platforms has achieved both performance improvements and cost savings.

6 Conclusions

In this research, we introduced the concept of an additive manufactured variable product platform and its associated process setting platform. Design and process adjustments based on a reference part were confined within a bounded FS in order to limit cost increments. A cost-driven design methodology was proposed to apply additive manufactured variable platform modules in product family design, aiming for performance improvement at lower costs. AM production costs were predicted using the proposed FTDABC approach, with time equations calculated from a trained ANFIS. The FS's boundary of AM process setting adjustment terms was identified by solving a multi-objective optimization problem that minimized cost increments and maximized design freedoms. A Mamdani model implemented with fuzzy design rules was used to calculate the extreme values of platform variants' design parameters, which were optimized to maximize customer perceived utilities in multiple market segments. Case studies on designing a family of R/C racing cars illustrated the proposed cost-driven variable platform design methodology. The proposed design methodology provided designers with a guideline to explore AM-enabled design space while controlling production costs at the same time. Compared to conventional product families with the drawback of performance compromise due to consistent platform sharing, the implementation of additive manufactured variable platforms demonstrated product performance improvements at lower costs.

Although the same general procedure of the proposed design methodology can be applied in different AM processes, definitions and mathematical expressions of process setting terms (e.g., Eqs. (4)–(8)) need to be uniquely formulated for each specific AM process. In future research, we will extend the methodology using other AM techniques to demonstrate its generalizability.

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnljTWeENb774MKM7HJ-nZUW6TxtbJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeQw by Universita Di Pisa user on 25 November

In this research, we calculated manufacturing costs per unit part using the FTDABC approach, without involving the production quantity in the equations. Therefore, the proposed platform design methodology is not able to include cost drivers such as procurement and warehousing, which are dependent on the production quantity. In future work, the effect of a product family's production quantity on variable platform designs will be investigated.

Acknowledgment

This research was supported by SERC, A*STAR Industrial Additive Manufacturing Programme, SIMTech-NTU Joint Lab, and an AcRF Tier 1 grant (RG94/13) from Ministry of Education, Singapore.

References

[1] Gibson, I., Rosen, D. W., and Stucker, B., 2010, *Additive Manufacturing Technologies*, Springer, New York.
[2] Bourell, D. L., Rosen, D. W., and Leu, M. C., 2014, "The Roadmap for Additive Manufacturing and Its Impact," 3D Print. Addit. Manuf., **1**(1), pp. 6–9.
[3] Simpson, T. W., Jiao, J., Siddique, Z., and Hölttä-Otto, K., 2014, *Advances in Product Family and Product Platform Design*, Springer, New York.
[4] Moon, S. K., Park, K., and Simpson, T., 2014, "Platform Design Variable Identification for a Product Family Using Multi-Objective Particle Swarm Optimization," Res. Eng. Des., **25**(2), pp. 95–108.
[5] Song, B., Dong, S., Deng, S., Liao, H., and Coddet, C., 2014, "Microstructure and Tensile Properties of Iron Parts Fabricated by Selective Laser Melting," Opt. Laser Technol., **56**(0), pp. 451–460.
[6] Cheng, W., Fuh, J. Y. H., Nee, A. Y. C., Wong, Y. S., Loh, H. T., and Miyazawa, T., 1995, "Multi-Objective Optimization of Part-Building Orientation in Stereolithography," Rapid Prototyping J., **1**(4), pp. 12–23.
[7] Canellidis, V., Giannatsis, J., and Dedoussis, V., 2009, "Genetic-Algorithm-Based Multi-Objective Optimization of the Build Orientation in Stereolithography," Int. J. Adv. Manuf. Technol., **45**(7–8), pp. 714–730.
[8] Strano, G., Hao, L., Everson, R. M., and Evans, K. E., 2011, "Multi-Objective Optimization of Selective Laser Sintering Processes for Surface Quality and Energy Saving," Proc. Inst. Mech. Eng., Part B, **225**(B9), pp. 1673–1682.
[9] Verma, A., and Rai, R., 2013, "Energy Efficient Modeling and Optimization of Additive Manufacturing Processes," Solid Freeform Fabrication Symposium, Austin, TX.
[10] Hopkinson, N., and Dickens, P., 2003, "Analysis of Rapid Manufacturing—Using Layer Manufacturing Processes for Production," Proc. Inst. Mech. Eng., Part C, **217**(1), pp. 31–39.
[11] Ruffo, M., Tuck, C., and Hague, R., 2006, "Cost Estimation for Rapid Manufacturing—Laser Sintering Production for Low to Medium Volumes," Proc. Inst. Mech. Eng., Part B, **220**(9), pp. 1417–1427.
[12] Atzeni, E., and Salmi, A., 2012, "Economics of Additive Manufacturing for End-Usable Metal Parts," Int. J. Adv. Manuf. Technol., **62**(9–12), pp. 1147–1155.
[13] Lindemann, C., Jahnke, U., Moi, M., and Koch, R., 2013, "Impact and Influence Factors of Additive Manufacturing on Product Lifecycle Costs," Solid Freeform Fabrication Symposium, Austin, TX.
[14] Lindemann, C., Jahnke, U., Moi, M., and Koch, R., 2012, "Analyzing Product Lifecycle Costs for a Better Understanding of Cost Drivers in Additive Manufacturing," Solid Freeform Fabrication Symposium, Austin, TX.
[15] Yim, S., and Rosen, D., 2012, "Build Time and Cost Models for Additive Manufacturing Process Selection," ASME Paper No. DETC2012-70940.
[16] Williams, C., Allen, J., Rosen, D. W., and Mistree, F., 2006, "Process Parameter Platform Design to Manage Workstation Capacity," *Product Platform and Product Family Design*, T. Simpson, Z. Siddique, and R. J. Jiao, eds., Springer, New York, pp. 421–455.
[17] Luo, X., Yang, W., Kwong, C., Tang, J., and Tang, J., 2014, "Linear Programming Embedded Genetic Algorithm for Product Family Design Optimization With Maximizing Imprecise Part-Worth Utility Function," Concurrent Eng., **22**(4), pp. 309–319.
[18] Weck, O. L. D., Suh, E. S., and Chang, D., 2003, "Product Family and Platform Portfolio Optimization," ASME Paper No. DETC03/DAC-48721.
[19] Chowdhury, S., Messac, A., and Khire, R. A., 2011, "Comprehensive Product Platform Planning (CP3) Framework," ASME J. Mech. Des., **133**(10), p. 101004.
[20] Jiao, J., Zhang, Y., and Wang, Y., 2007, "A Generic Genetic Algorithm for Product Family Design," J. Intell. Manuf., **18**(2), pp. 233–247.
[21] Suh, E., De Weck, O., and Chang, D., 2007, "Flexible Product Platforms: Framework and Case Study," Res. Eng. Des., **18**(2), pp. 67–89.
[22] Wang, L., Song, B., Li, X., and Ng, W. K., 2007, "A Product Family Based Life Cycle Cost Model for Part Variety and Change Analysis," International Conference on Engineering Design (ICED'07), Paris, France, Paper No. DS14_P_152.
[23] Bryan, A., Wang, H., and Abell, J., 2013, "Concurrent Design of Product Families and Reconfigurable Assembly Systems," ASME J. Mech. Des., **135**(5), p. 051001.
[24] Simpson, T., Siddique, Z., and Jiao, J., 2006, "Platform-Based Product Family Development," *Product Platform and Product Family Design*, T. Simpson, Z. Siddique, and R. J. Jiao, eds., Springer, New York, pp. 1–15.
[25] Kaplan, R. S., and Anderson, S. R., 2004, "Time-Driven Activity-Based Costing," Harv. Bus. Rev., **82**(11), pp. 131–140.
[26] Chansaad, A., Rattanamanee, W., Chaiprapat, A., and Yenradee, P., 2012, "Fuzzy Time-Driven Activity-Based Costing Model in an Uncertain Manufacturing Environment," Asia Pacific Industrial Engineering and Management Systems Conference, Phuket, Thailand.
[27] Sarokolaei, M. A., Saviz, M., Moradloo, M. F., and Dahaj, N. S., 2013, "Time Driven Activity Based Costing by Using Fuzzy Logics," Procedia–Soc. Behav. Sci., **75**(0), pp. 338–345.
[28] Dadbakhsh, S., Hao, L., Jerrard, P. G. E., and Zhang, D. Z., 2012, "Experimental Investigation on Selective Laser Melting Behaviour and Processing Windows of In Situ Reacted Al/Fe_2O_3 Powder Mixture," Powder Technol., **231**(0), pp. 112–121.
[29] Leekwijck, W. V., and Kerre, E. E., 1999, "Defuzzification: Criteria and Classification," Fuzzy Sets Syst., **108**(2), pp. 159–178.
[30] Jang, J. S. R., 1993, "Anfis: Adaptive-Network-Based Fuzzy Inference System," IEEE Trans. Syst., Man Cybernetics, **23**(3), pp. 665–685.
[31] Verma, A., and Rai, R., 2014, "Computational Geometric Solutions for Efficient Additive Manufacturing Process Planning," ASME Paper No. DETC2014-34067.
[32] Babuška, R., 2003, "Neuro-Fuzzy Methods for Modeling and Identification," *Recent Advances in Intelligent Paradigms and Applications*, A. Abraham, L. C. Jain, and J. Kacprzyk, eds., Physica-Verlag HD, Heidelberg, Germany, pp. 161–186.
[33] "Willit3dprint," last accessed July 13, 2015, http://www.willit3dprint.com/
[34] Konak, A., Coit, D. W., and Smith, A. E., 2006, "Multi-Objective Optimization Using Genetic Algorithms: A Tutorial," Reliab., Eng. Syst. Saf., **91**(9), pp. 992–1007.
[35] Hague, R., Mansour, S., and Saleh, N., 2004, "Material and Design Considerations for Rapid Manufacturing," Int. J. Prod. Res., **42**(22), pp. 4691–4708.
[36] Samperi, M., Chernow, E., Simpson, T. W., Joshi, S., and Talbot, M., 2013, "Towards a Process Workflow for Designing and Fabricating Parts Using Additive Manufacturing," 2013 RAPID Conference and Exposition, Pittsburgh, PA, June 10–13, Paper No. 64832.
[37] Vayre, B., Vignat, F., and Villeneuve, F., 2012, "Designing for Additive Manufacturing," Procedia CIRP, **3**(0), pp. 632–637.
[38] Vayre, B., Vignat, F., and Villeneuve, F., 2013, "Identification on Some Design Key Parameters for Additive Manufacturing: Application on Electron Beam Melting," Procedia CIRP, **7**(0), pp. 264–269.
[39] Train, K. E., 2009, *Discrete Choice Methods With Simulation*, Cambridge University Press, Cambridge, UK.
[40] "Traxxas Products," last accessed Jan. 22, 2015, http://traxxas.com/products
[41] Rosen, D. W., 2014, "Research Supporting Principles for Design for Additive Manufacturing," Virtual Phys. Prototyping, **9**(4), pp. 225–232.
[42] Moon, S. K., Tan, Y., Hwang, J., and Yoon, Y.-J., 2014, "Application of 3D Printing Technology for Designing Light-Weight Unmanned Aerial Vehicle Wing Structures," Int. J. Precis. Eng. Manuf.-Green Technol., **1**(3), pp. 223–228.

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/138/4/041701/6228413/md_138_04_041701.pdf?casa_token=GgNE-h8umMMAAAAA:HOVq5TTPnljTWeENb774MKM7HJ-nZUW6TxtdJZZdaGp1CxXRRWvJ6dh1kp4UW2sMfbeQw by Universita Di Pisa user on 25 November